No of Pages : 3 **Course Code : 23MXAM**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004
SEMESTER EXAMINATIONS, JUNE 2024
MCA Semester : 2
23MXAM OPEN SOURCE SYSTEMS

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.
3. Course Outcome : Table

Qn.1	CO1.	Qn.2	CO.2	Qn.3	CO3	Qn.4	CO.4

COs	At the end of learning this course, students will be able to
CO1	understand the concepts and philosophy of Open Source Systems and analyse issues and significance of software licensing and patenting.
CO2	understand and apply scripting languages for solving real world problems.
CO3	analyze the features of NoSQL DB and apply proper Query and index operations to solve the given problem.
CO4	understand and analyze the features, working and license of various open source software tools.

		Question	(BTL)
1	a)	Defend the social impacts of open-source systems beyond technology.	(L5)
	b) i)	Analyze any three implications of copyleft license by the Linux kernel, licensed with General Public License version 2 (GPLv2). (3)	(L4)
	ii)	Discuss towhat extent the bazaar model promotes knowledge sharing, cross-functional collaboration, and collective problem-solving within development communities over the cathedral model. (7)	(L6)
	c)	Analyze the steps involved in bug life cycle in software testing with emphasis on different status of bugs.	(L4)
2	a)	Write about any 3 key features of R that make it a popular choice for application developers.	(L2)
	b) i)	Recommend proper R code to create a matrix that stores order details, including OrderID, ProductID, UnitPrice, and Quantity, as follows:	(L5)

OrderID	ProductID	Unit Price	Quantity
1001	101	55	3
1002	203	520	5
1003	213	210	3

Also determine properR script to find the total price of each order, the total number of items sold, and the total cost of all orders. (7)

ii) Analyze the following code and find the output or error. (L4)

```
theList=list(order_matrix[,3],order_matrix[,4])
print(theList)
l=lapply(theList,sum)
print(l[2])
```

Assume the order_matrix is the one created in the q 2)b)i) (3)

c) Examine the minimal arguments needed to read and write a csv using R language. (L4)

Assuming the following table "orders" is stored as sample.csv file, develop proper R script to

i) read the csv file and store it as data frame.

ii) update the BloodPressure for Sub004 to 130.5

ii) modify the Category as 2 for individuals with ages between 16 to 18

iii) calculate the average age of all individuals

iv) modify the Aneurisms_q1 for females with Category= 2 to 120

v) store the updated data back to the sample.csv file.

Table: orders

ID	Gender	Category	BP_mm Hg	Age	Aneurisms _q1_mm	Aneurisms_ q2_mm
Sub001	M	3	132	16.0	114	140
Sub002	M	2	139	27.2	148	209
Sub003	F	3	130	29.5	196	251
Sub004	F	1	105	15.7	199	140
Sub005	M	1	125	19.9	188	120
Sub006	M	2	112	14.3	260	266

3 a) State and define three main benefits of NoSQL over SQL databases. (L1)

b) i) In a Facebook-like social media platform, users post updates and comments, and these updates need to be visible all the time to users in real-time. If Facebook wants to ensure strict consistency across all nodes it faces heavy latency losing users interest. Justify how a replicated Mongo DB could keep users intact by prioritizing availability and allowing eventual consistency in case of network partition. (3) (L5)

ii) Imagine there is an e-commerce platform that sells various products online. It uses a Mongo DB collection named orders that stores information about customer orders, including details such as order ID, customer ID, product ID, quantity, price, and order date. Additionally, you have a separate collection named products that contains information about each product, such as product ID, name, category, and price. (L3)

Sample documents in "orders" collection are:

```
[ { "_id": ObjectId("60b9d9f178e82c359c82ae23"),
  "order_id": "1001", "customer_id": "C001",
"product_id": "P001",
  "quantity": 2, "order_date": ISODate("2023-08-
15T10:30:00Z") },
 { "_id": ObjectId("60b9d9f178e82c359c82ae24"),
  "order_id": "1002", "customer_id": "C002",
"product_id": "P001",
  "quantity": 4, "order_date": ISODate("2023-08-
16T09:45:00Z") },
{ "_id": ObjectId("60b9d9f178e82c359c82ae23"),
  "order_id": "1001", "customer_id": "C001",
```

```
"product_id": "P002",
   "quantity": 5, "order_date": ISODate("2023-09-15T10:30:00Z") },
{"_id": ObjectId("60b9d9f178e82c359c82ae24"),
   "order_id": "1002", "customer_id": "C003", "product_id": "P001",
   "quantity": 14, "order_date": ISODate("2023-08-16T12:45:00Z") }, … ]
```

Sample documents in “products” collection are:

```
[ { "_id": ObjectId("60b9d9f178e82c359c82ae01"), "product_id": "P001",
   "name": "Smartphone", "category": "Electronics", "price": 50000 },
  { "_id": ObjectId("60b9d9f178e82c359c82ae02"), "product_id": "P002",
  "name": "Laptop", "category": "Electronics", "price": 80000 }, … ]
```

Identify proper queries to determine

i) total revenue generated by product category

ii) top 5 best selling products

iii) top customer who repeatedly places order

iv) the total selling cost raised for each order.

v) top 3 customers who have purchased more products (7)

c) How does two-Phase Commit protocol ensure transactional integrity and session consistency in distributed NoSQL databases? Analyze how a two phase commit process could help in real-time consistent interactions and display of the consistent object move and all other actions in a distributed online gaming platform. (L4)

4. a) What is a web content Management system? How does Joomla support it? (L1)

b) i) Compare the web frameworks Rails and Bootstrap on their use for developers, architecture and functionality (3) (L5)

ii) Unlike Bootstrap framework, Rails provides built-in security features. Describe the security features provided by Rails. (7) (L6)

c) i) Elaborate on the key advantages of using Selenium for automated testing compared to manual testing. Also discuss the effects of Selenium's open-source license Apache License. (L4)

(OR)

ii) How Git works as a distributed version control system? Analyze on the different storage areas Git uses to save the code before and after commit.

CO : COURSE OUTCOME; BTL : BLOOM’S TAXONOMY LEVEL

/END/

FD/RL